

# Hybrid Quantum Cryptanalysis of Grain-128AEAD using Grover's Algorithm: An Analysis of Oracle Selectivity

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## 1. Abstract

This paper explores the quantum security of the Grain-128AEAD stream cipher using a hybrid cryptanalysis approach that combines classical cipher implementation with quantum search techniques. A full bit-accurate implementation of Grain-128AEAD is developed, and a Grover-based quantum attack is simulated using Qiskit.

Due to the infeasibility of constructing a fully reversible quantum circuit for the cipher, a hybrid oracle model is employed based on partial ciphertext matching. The study investigates the impact of oracle selectivity on the effectiveness of Grover's algorithm by varying the number of matched output bits.

Experimental results demonstrate a clear transition from distributed probability distributions under weak oracle conditions to deterministic amplification in the case of a uniquely identifying oracle. This behavior is consistent with theoretical predictions of Grover's algorithm, where amplification depends on the ratio of total states to valid solutions.

The findings highlight the critical role of oracle design in quantum cryptanalysis and provide insights into how partial information leakage can influence the feasibility of quantum key search attacks. This work serves as a practical framework for analyzing the quantum resistance of stream ciphers under realistic constraints.

## RESULTS AND DISCUSSION

### 6. Results and Discussion

#### 6.1 Experimental Setup

The experiments were conducted using a hybrid framework combining a classical implementation of the Grain-128AEAD cipher with a quantum simulation of Grover's algorithm using Qiskit.

A reduced key space of 8 bits was used to enable efficient simulation, while preserving the structural behavior of quantum search. The oracle was constructed using partial matching of ciphertext bits derived from the classical encryption process.

Grover's algorithm was executed with varying oracle strengths by selecting different numbers of target bits (5, 7, and 8). The number of iterations was chosen based on the theoretical estimate:

$$k \approx (\pi/4) \sqrt{N / M}$$

where :

\*)  $N = (2^n)$  is the search space

\*)  $M$  is the number of valid solutions

## 6.2 Effect of Oracle Selectivity

The primary objective of the experiment was to study how the number of matched output bits (oracle selectivity) affects the amplification behavior of Grover's algorithm.

Case 1: Weak Oracle (5-bit Matching)

1.) Approximate solutions:

$$M \approx (2^{(8-5)}) = 8$$

2.) Observation:

a.) The probability distribution remained relatively uniform

b.) No clear dominant state was observed

Interpretation:

When multiple candidate keys satisfy the oracle condition, Grover's algorithm amplifies all valid states simultaneously, resulting in a distributed probability landscape.

Case 2: Moderate Oracle (7-bit Matching)

1.) Approximate solutions:

$$M \approx (2^{(8-7)}) = 2$$

2.) Observation:

a.) Two dominant peaks emerged

b.) Significant amplification compared to other states

Interpretation:

Reducing the number of valid solutions increases amplification strength. Grover's algorithm concentrates probability mass over fewer states, leading to distinguishable peaks.

Case 3: Strong Oracle (8-bit Matching)

1.) Approximate solutions:

$$M \approx 1$$

2.) Observation:

a.) A single state dominated all measurements

b.) 100% success rate (4000/4000 shots)

Interpretation:

This represents the ideal case where the oracle uniquely identifies the correct key. Grover's algorithm achieves near-deterministic behavior, fully amplifying the correct solution.

## 6.3 Relationship Between Oracle Strength and Amplification

The results demonstrate a clear relationship:

Increasing oracle selectivity → Decreasing solution space → Stronger amplification

This aligns with Grover's theoretical behavior, where the probability of measuring the correct state increases as the number of valid solutions decreases.

## 6.4 Impact of Grover Iterations

The number of Grover iterations plays a critical role in amplification.

- a.) For smaller solution sets, fewer iterations are required
- b.) Exceeding the optimal number leads to over-rotation, reducing success probability

This behavior follows the oscillatory nature of Grover's algorithm, where probability amplitudes rotate in the solution space.

## 6.5 Circuit Complexity Analysis

The quantum circuit analysis yielded the following:

- a.) Qubits: 9
- b.) Depth: 122
- c.) Total gates: 546
- d.) Multi-controlled gates (MCX): 24

Although the circuit reports zero Toffoli gates, this is due to abstraction in Qiskit. In practice, each MCX gate decomposes into multiple Toffoli gates, contributing significantly to the overall quantum cost.

Thus, the actual resource requirements are substantially higher than indicated by the raw gate count.

## 6.6 Limitations

The following limitations should be noted:

- 1.) The oracle is constructed using partial ciphertext matching rather than full reversible cipher computation
- 2.) The key size is reduced (8-bit) for simulation feasibility
- 3.) The 8-bit matching case represents an idealized scenario, serving as an upper bound on Grover's performance

## 6.7 Key Insight

The most significant outcome of this work is:

The effectiveness of Grover's algorithm in cryptanalysis is strongly dependent on oracle selectivity. As the oracle becomes more restrictive, the solution space reduces exponentially, leading to sharper and eventually deterministic amplification.

## 7. Conclusion

In this work, a hybrid framework was developed to study the behavior of Grover's algorithm in the context of stream cipher cryptanalysis. A classical implementation of Grain-128AEAD was combined with a quantum search model to simulate key recovery under varying oracle conditions.

The results demonstrate that oracle selectivity plays a crucial role in determining the effectiveness of Grover's algorithm. As the number of matched output bits increases, the solution space reduces exponentially, leading to stronger amplitude amplification and improved success probability. In the ideal case of a uniquely identifying oracle, the algorithm achieves near-deterministic outcomes.

However, such ideal conditions are not typically available in real-world cryptanalysis, where only partial information can be exploited. The hybrid approach adopted in this work provides a realistic and scalable method for studying quantum attacks without requiring full reversible implementations of complex ciphers.

**Overall, this study emphasizes that the practical impact of quantum attacks depends not only on algorithmic speedups but also on the structure and strength of the oracle. Future work will focus on constructing reversible reduced-round versions of Grain and performing detailed quantum resource estimation to better understand its resistance against large-scale quantum adversaries.**

**NOTE : Paper is still in progress.**